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(57) Abstract :
ABSTRACT The present disclosure relates to an innovative system and method for enhancing the functionality and efficiency of public toilets through the integration of Internet of Things (IoT) technologies. The system addresses the existing challenges associated with maintaining cleanliness, facilitating timely maintenance, and enabling effective monitoring in public sanitation facilities. By deploying a network of interconnected IoT sensors and devices, the proposed invention facilitates the automated monitoring of critical parameters, including ingress and egress counts, occupancy status, ambient air quality, smoke detection, and sanitation levels. The system further enables real-time data acquisition and transmission, thereby minimizing the need for manual inspections and optimizing resource allocation for maintenance and upkeep. Furthermore, the invention incorporates an automated disinfectant dispensing mechanism to proactively mitigate the spread of contagious diseases and elevate overall hygiene standards within the public toilet environment. This IoT-enabled solution provides a cost-effective and highly efficient approach to ensuring and sustaining public toilet cleanliness, contributing to improved public health outcomes while simultaneously reducing operational expenditures and manpower requirements.

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